# 4. Instrumentation

Introduction

Radiation detection instruments monitor the effect of ionizing radiation on a given material. For example, the ions produced in a given volume of gas can be measured, and the ions will be proportional to the energy absorbed by the gas. No single instrument performs acceptably under all conditions and requirements. Therefore, no single instrument can monitor and discriminate between all types of radiation at the various energy levels. In this chapter, the most commonly used radiation detectors will be discussed, including:

Ionization chambers
Proportional counters
Geiger-Mueller (G-M) counters
Scintillation detectors
Photographic devices
Solid-state and activation devices

Specific applications of these detectors in personnel monitoring devices, including film badges, pocket dosimeters, and pocket ion chambers, will also be presented.

The obvious purpose of monitoring radioactivity in the working environment is to ensure that the exposure rate and absorbed dose for any given individual are less than the maximum permissible dose (MPD) equivalents established as a minimum standard. However, the response recorded on a detector is due to the energy absorbed by the detector itself; and in some instances the detector does not have the same properties as human tissue. It is possible to select a detector and probe that can provide readings proportional to the absorbed dose that would be experienced by an employee in a given environment. Generally, however, the response of the instruments is not exactly equivalent to the response of human tissue receiving the same amount of energy. This is due primarily to the secondary ionization that would normally occur in human tissue but does not necessarily occur in the detector. For purposes of this discussion, however, it may be assumed that the response of the detector is equivalent to the absorbed dose unless otherwise stated.

The absorbed dose for an individual can rarely be determined directly, if at all. An estimation made by using some type of detector must be acceptable. Realizing the absorbed dose cannot be determined and that an estimation must be used, the maximum permissible dose equivalent values that have been previously presented take into consideration that an estimation will be used and, therefore, MPD values are conservative. These values have been

conservatively established to the extent that an error in actual reading of ±30% could be tolerated. In most instances, the instruments selected, if appropriately matched to the environment and type of radiation of interest, will fall within this range. In general, the accuracy of the detectors for photon energies tends to be somewhat better than for particulate radiation.

If the detection instruments are calibrated according to the manufacturer's recommendation for the environment and radiation type of interest, no further effort is usually necessary to increase the accuracy of the instrument. However, if the readings should approximate the maximum permissible dose equivalent established, it is important that the most accurate readings possible be obtained. If the MPD value is approximated, the individual receiving this exposure may receive different medical treatment based upon the amount of exposure that has been calculated. Therefore, to ensure appropriate treatment, the most accurate reading possible should be obtained. For readings that are well above or well below the MPD value, the need for accuracy is not as great.

## Instrumentation

This section deals with the most common radiation detection and monitoring devices presently being used.

Ionization Chamber Instruments. The ionization chamber is an instrument that monitors the change in a gas caused by ionizing radiation. Ionizing radiation falls on the gas contained in the chamber and forms ions that migrate toward the cathode and anode within the chamber and are collected there. The potential voltage difference that is created because of the migration of the primary ions is measured. This potential difference is proportional to the quantity of ionizing radiation that falls upon the chamber.

This instrument can be used to measure relatively high levels of radiation. It can measure both particulate and electromagnetic radiation. The ionization chamber responds to any ionization produced in the chamber itself. It does not have the capability of discriminating between radiation types and specifically between particles of different linear energy transfer (alpha versus beta).

The ionization chamber is usually 190 to 320 cubic centimeters in volume. In general, the larger the chamber, the greater the sensitivity and required operational voltage. It can be designed to measure all types of radiation. The chamber is usually open to the atmosphere and requires corrections for ambient temperature and pressure. Sealed chambers are also available; they do not require corrections for the ambient temperature and pressure but may have reading changes because of leakage or absorption and adsorption on inside surfaces of the gas within the chamber. The thickness of the wall of the chamber is of critical importance. The thickness must approximate the maximum range of the ionized particles produced by the ionizing radiation. Table 6.4.1 illustrates the various wall thicknesses required, based upon the energy of the radiation.

The chambers can also be designed to be air or tissue equivalent. If the chamber is designed to be air equivalent, then the readings reflect exposure rates. If the chambers are designed to be tissue equivalent, then the properties of the chamber simulate human tissue and the readings are calibrated to measure absorbed dose.

Table 6.4.1

Thicknesses of ionization chamber walls required for establishment of electronic equilibrium (ICRU, 1964).

| Photon energy (MeV) | Thickness ($g \cdot cm^{-2}$) |
|---|---|
| 0.02 | 0.0008 |
| 0.05 | 0.0042 |
| 0.1 | 0.014 |
| 0.2 | 0.044 |
| 0.5 | 0.17 |
| 1 | 0.43 |
| 2 | 0.96 |
| 5 | 2.5 |
| 10 | 4.9 |

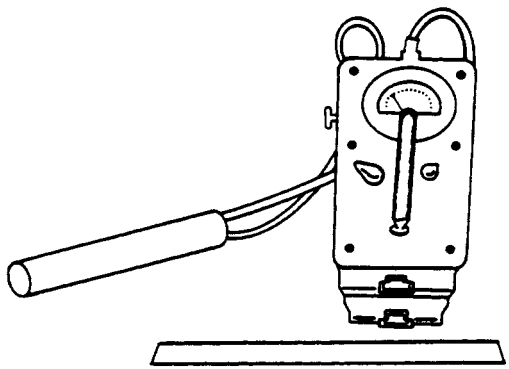
The primary advantage of the ionization chamber instrument is that it is a simple, rugged device that reliably determines absorbed dose or exposure. The disadvantage of this instrument is that special designs or modifications are required for the instrument to discriminate well between the types of radiation. Because of this, it is necessary to have a general knowledge of the radiation spectrum being evaluated.

Proportional Counter Instruments. THe proportional counter (Figure 6.4.1) functions on the same principle as the ionization chamber. However, the voltage on the collection plates is increased such that as the primary ions approach the collection plates, secondary ionization occurs. This secondary ionization and the ions produced from it contribute to the ion current pulse that is registered on the meter. This process is referred to as "gas amplification." This gas amplification process increases the sensitivity of the proportional counter over a simple ionization chamber by a factor of $10^3$ to $10^4$ times. Even though gas amplification does occur, there still remains a proportionality between the counter current and energy fluence rate of the radiation which enters the proportional counter.

The proportional counter is more useful in measuring particle radiation than photon radiation. It has the capability of measuring both alpha and beta radiation and can discriminate between the two types if they are simultaneously present. If the chamber is lined with a boron film or filled

Figure 6.4.1

Proportional counter.



with $BF_3$ gas, then thermal neutrons can also be monitored. Fast neutrons of low energy level may also be measured with this instrument. Because of the proportionality between the counter and the energy entering the chamber, it is possible to use the proportional counter for spectrometry.

The selection of the type of gas and wall construction for the proportional counter chamber is dependent upon the purpose and type of measurements that are to be made. As with the ionization chamber, the proportional counter chamber can be designed to be air or tissue equivalent. Further, by changing the thickness of the wall, one can get an indication of the absorbed dose at various levels in human tissue.

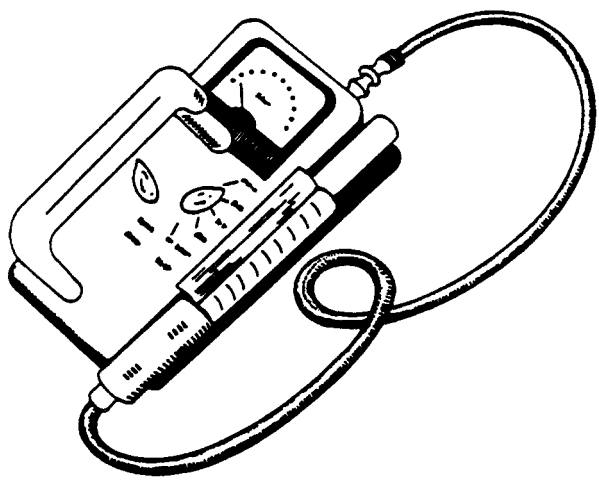
The primary advantages of the proportional counter are its discrimination capabilities. For example, it is possible for the counter to eliminate the effects of gamma radiation while counting neutrons and to negate the effects of beta when counting alpha. Its primary advantage is its high sensitivity with relative high accuracy and counting efficiency. The disadvantages of this unit include the insulation requirements around the components to prevent miscellaneous radiation from entering the components. Also, in some instances, inaccuracies in the readings may occur due to losses of absorbed energy. These losses may decrease accuracy by as much as 50%.

Geiger-Mueller (G-M) Counter. The Geiger-Mueller counter (Figure 6.4.2) is used extensively as a sensitive radiation detector. The G-M counter operates much like the ionization chamber and proportional chamber, with the exception that the applied voltage is increased still further so that all gas atoms in the chamber undergo secondary ionization. Because of the extensive secondary ionization, the sensitivity is greatly increased. However, the electrical pulse current generated internally is not generally proportional to the type of ionizing event, and it cannot be directly related to either absorbed dose or exposure. Therefore, the G-M counter is used primarily as a

detector of potential radiation hazard but cannot usually be used as a monitoring device.

Figure 6.4.2

Geiger-Mueller counter.



The G-M counter is used primarily to detect the existence of low-level radiation, especially in the range of 0-20 milliroentgens per hour. Under special conditions, the G-M counter can be modified to yield a response that is proportional to the exposure over a limited range of photon energies, but primarily it is used as a detection device for low-energy beta and gamma radiation. The G-M counter has the capability of monitoring gamma radiation in the presence of neutrons.

The G-M counter is made in a variety of shapes, sizes, and compositions. These factors must be matched to the environment and the type of radiation that is to be monitored. In some instances, the sensitivity to beta radiation is dependent upon the angle of incidence of radiation; that is, the G-M counter is direction-dependent. Survey procedures that are used to detect radiation must take into consideration the directional factor of the G-M counter.

One characteristic of the G-M counter that also must be considered is the so-called "dead time." Dead time, which ranges from approximately ten to several hundred microseconds, is the time required for the ion avalanche to be initiated, to travel along the anode, and to be quenched, leaving the counter ready for a new count. Because of the G-M counter designs and the massive secondary ionization caused by ionizing radiation and high voltage, time is required for the instrument to respond to the ionization in the chamber,

initiate the avalanche, and then have the chamber return to a neutral state so that ionization may again occur. The halting of the avalanche is accomplished by the inclusion of an inert gas in the ion chamber. This gas is commonly referred to as a "quenching gas."

The G-M counter should not be used if it is anticipated that the count rate will be greater than 1000 cpm (counts per minute). Because the G-M counter is energy dependent, an increased radiation level will cause a decrease or blockage of the meter reading. This occurs because the mechanics of a G-M counter cannot operate with such a high level of ionizing radiation. In this instance, the G-M counter may show a "no field" or no ionizing radiation present when, in fact, the field is very high. In the instance where meter blockage occurs, no count will be registered. If the meter is not blocked, at least one count every few seconds will occur because of typical background radiation present in the environment. Therefore, if no counts register during the first few seconds, the counter is either inoperative or blocked. In this case, a less sensitive detector should be used, such as an ionization chamber. For the same reason, G-M counters should not be used for the measurement of short, high-intensity pulses since this may also cause meter blockage.

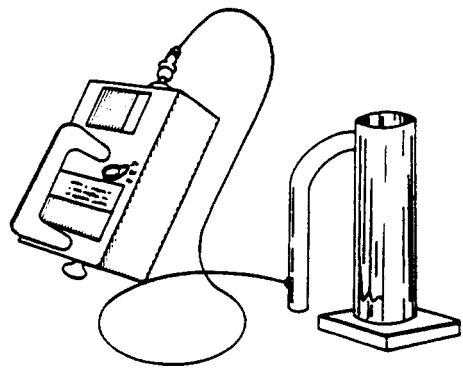
The primary advantage of the G-M counter is that the unit is typically portable, generally stable and rugged, and has a high sensitivity. The disadvantages of this unit are that the readings cannot be directly related to absorbed dose or exposure but merely indicate the presence of ionizing radiation. Further, because of the presence of a high radiation field, the meter may block and read "no field" at 1000 cnts/min. Finally, most G-M counters have a directional sensitivity, requiring a sweeping-type motion during any survey process.

Scintillation Detector. The scintillation detector (Figure 6.4.3) works on a different principle than the ionization chamber instruments previously discussed. With the scintillation detector, the ionizing radiation interacts with a phosphor or crystal that is capable of producing light. When the ionizing radiation falls on the scintillation counter, the crystal is excited and emits light. The light produced registers on a photomultiplier and is converted to electrical impulses. The electrical impulses are then magnified and registered on a microammeter. The number and size of the electrical impulses recorded are related to the energy deposited in the scintillator chamber.

The scintillation detector has basically three components. The scintillator itself is a photo-sensitive crystal, plastic, or liquid that, when exposed to ionizing radiation, will emit light. The photomultiplier collects the light and converts it to electrical impulses. The final component of the system is the electronic counting equipment that registers and records the electrical current produced. The scintillation detector is most useful in the evaluation of the alpha and low level gamma radiation. With a sodium iodide crystal, gamma radiation may be measured in the presence of beta. Also, the instrument may be modified so that beta radiation may be measured in the presence of high-energy gamma. Further, soft X-rays may also

Figure 6.4.3

Scintillation-type instrument.



be monitored using this instrument. Alpha particles using a cesium iodide crystal can be evaluated. To avoid the problem of the window of the chamber absorbing alpha particles, the radioactive sample can be placed in the scintillation chamber and a direct reading taken without interference from the window.

The primary advantage of this type of instrument is its sensitivity. It is possible to monitor low-energy sources in the range of 5 μR/hr above background radiation. It can discriminate between different types and energies of radiation and has a relatively high counting efficiency; i.e., alpha and beta particle monitoring approached 100% efficiency. Finally, the unit can be used as a spectrometer so that a mixture of several radionuclides can be analyzed quantitatively and qualitatively. This can be done because of the direct relationship between the pulse shape and height and the type and energy of the radiation being monitored.

One problem with this unit is that the photomultiplier and crystals are relatively fragile. The unit can be designed as a portable unit, but care must be taken not to damage the internal parts of the unit. Further, some crystals used are easily damaged by moisture and humidity and will cause erroneous readings because of environmental conditions. Care must be taken also to keep the scintillator and photomultiplier in a light-tight case to ensure that erroneous readings are not taken. The instrument is usually designed to provide this light-tight case. However, because of wear and use, extraneous light can enter the system and cause erroneous readings.

Photographic Devices. Photographic capabilities can also be used to monitor ionizing radiation. Photographic devices can be designed to provide a reasonably accurate and permanent record of cumulative exposure. Photographic films are available that are sensitive to the various types of ionizing

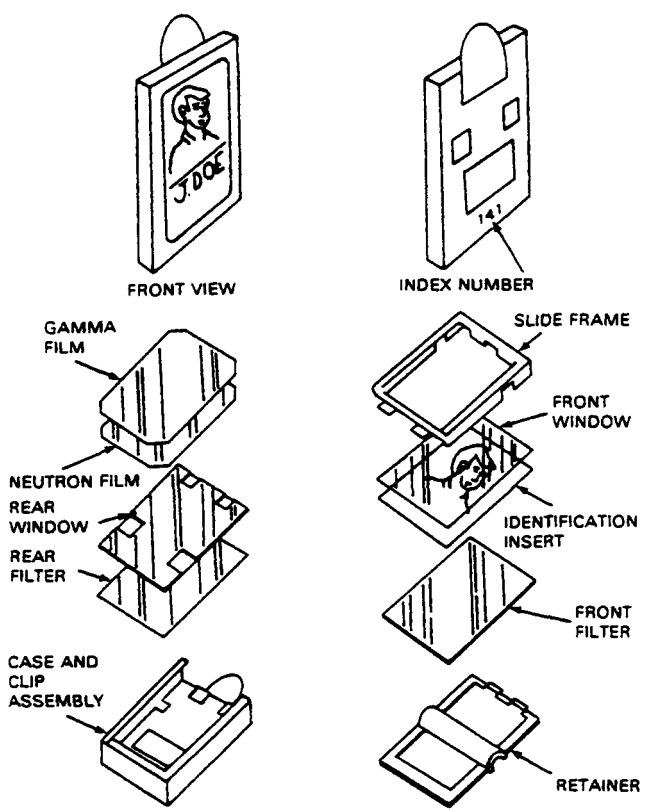
radiation. The ionizing radiation interacts with the silver halide in the photographic emulsion on the film. The silver is ionized and then attracted to negatively charged sensitivity centers on the crystals in the film emulsion. At this point, the silver ions produced are reduced to a free silver compound. When the film is processed, the silver ions are removed; and the free silver stays attached to the film. The quantity of silver that remains is proportional to the exposure of ionizing radiation.

The range of the film and the type of radiation to which it is sensitive are dependent upon the characteristics of the emulsion, the filtration that might be used, the processing technique, and the type and quality of exposing radiation. In general, the range of the emulsion is usually 10:1.

The primary use of the photographic device is in personnel monitors. These are commonly used as "film badges." Figure 6.4.4 is an illustration of a film badge.

Figure 6.4.4

Radiation film badge.



Film badges generally are used to monitor beta, neutron, X-, and gamma radiation. Because the film must be maintained in a light-tight casement, only beta energy of greater than 0.2 MeV can penetrate the film casement. This also eliminates the possibility of using the film badge to monitor alpha radiation.

Usually the film badges are quite small. They require no power supply to operate, and special holders and packs have been devised to monitor various parts of the body; e.g., wrists and fingers. In many instances, larger film casements and film processes have been used to monitor ionizing radiation.

Using the photographic devices as a monitoring process is relatively expensive and is not generally cost effective for less than two hundred people. Usually film badges are changed every two to four weeks and then processed. However, in some areas if radiation is either very critical or noncritical, they may be changed weekly or quarterly, as the system requires. A potential problem in using the film badge and not changing the badge on a biweekly or monthly basis is that high levels of accidental exposure may be undetected for long periods of time; e.g., if the film badges are changed on a quarterly basis.

Use of the film badge provides some distinct advantages. It provides a permanent record of radiation exposure and can be used on large populations of individuals. Film badges can be designed to discriminate between types of radiation and, because of their size and stability, can be used to monitor whole-body as well as individual parts of the body: e.g., fingers. However, there are some problems with using the film badge that should be considered. First, the film badge and exposure of ionizing radiation to the film badge is dependent upon the direction of the incident radiation. Therefore, if the film badge is not appropriately placed with respect to the potential source and the individual receiving the exposure, the readings on the film badge will be incorrect. Second, it is difficult to assess the absorbed dose and dose equivalent to within 20 to 50 percent for simple radiation fields. In those areas where a complex or multi-course radiation field may exist, the error may be even greater. Third, calibration and processing of the film is somewhat complex. Because of the variations in the reactivity of the film emulsions from batch to batch, it is necessary to expose some of the film to a standardized radiation source and use this as the standard for that particular batch. This variation in sensitivity of the emulsions can cause problems in interpretation and calibration of the film system.

Finally, some film emulsions are reactive to water vapor, causing potential error in readings. Therefore, it is necessary to carefully contain all film badges in moisture-tight casements.

Solid-State and Activation Devices. Any solid material that, when radiated, exhibits a property whose response is a function of the energy absorbed may in principle be used as a radiation device. Aside from the properties already discussed, other physical properties that have been considered include coloration, photoluminescence, thermoluminescence, and photoconductivity. Of these, thermoluminescent, photoluminescent, and semiconductor detectors are also in common use.

Thermoluminescent detectors are useful in measuring X- and gamma radiation and high-energy beta particles. The thermoluminescent material, when exposed to radiation, is ionized and "holes" develop in the lattice structure of the material. In essence, these "holes" are areas in the lattice where an electron has been removed by the ionizing radiation. When the material is heated, this provides adequate energy for the electrons to shift in the material and fill the holes. As this recombination occurs, light is emitted. The quantity of light emitted is proportional to the ionizing radiation exposure that originally caused the change in the lattice structure. Typically, materials such as calcium chloride, lithium chloride, and calcium phosphate are used for thermoluminescent detectors.

The advantage of these detectors is that they can be quite small, e.g., 0.1-1 gram. They are rugged and durable and provide monitoring over a wide range (10 milliroentgens, $10^5$ roentgens). The reading provided usually is within $\pm$5-10 percent, and the detector is reusable. The disadvantage of this type of detector is that an instrument is required to "read" the device. This reading device heats the detector and then records the light emitted. One problem that does exist with the use of thermoluminescent detectors is that of fading. The heat at room temperature may cause recombination and thus a reduction in the signal. For example, calcium sulphate readings may decrease by as much as 30 percent after eight hours and as much as 65 percent after eight days. Because of this problem, it is necessary to read the detector as soon as possible after potential exposure.

Photoluminescent devices work on a similar principle as thermoluminescent devices, with the exception that rather than exposing the detector to heat to cause the emission of light, the photoluminescent device is exposed to ultraviolet light, thus causing light emission. Photoluminescent devices have the same advantages as the thermoluminescent devices including the long range and portability. They also share common problems in that a device is required to read the detector and fading may occur.

One final type of device that may be encountered is a device based upon the semiconductor principle. With this device, an exposure to ionizing radiation causes a change in conductivity of the subject material. This change in conductivity is then measured and, in some instances, is proportional to the exposure dose. This type of device can be used to measure alpha, beta, X-, or gamma radiation. In general, it acts as a solid ionization chamber with the exception that the range is from $10^{-6}$ to $10^4$ roentgens per hour. The major problem with this type of device is that the semiconductor may be temperature sensitive.

## Personnel Monitoring Devices

Because of the importance of personnel monitoring, this section reviews the major types of detection devices used for personnel monitoring.

Film Badges. Film badges were previously discussed. As mentioned they are designed to provide a reasonably accurate and permanent record of cumulative exposure. They can be used to monitor whole-body or parts of the

body as required. The primary problem as mentioned with the film badges is the complexity and time required for film processing.

Pocket Dosimeter. The pocket dosimeter is widely used in radiation monitoring. The dosimeter works on the principle of an ion chamber. It has the capability of indicating accumulated exposure to radiation at any given time and can be read by the individual without any type of reading instrument. However, the pocket dosimeter does not provide a permanent record of exposure; but it may be recharged which will reset the monitor to zero for reuse. The pocket dosimeter is prepared with a radiation scale to a maximum of 200 milliroentgens.

Pocket Chamber. The pocket chamber is also commonly used in monitoring radiation. It is similar in size and shape to a fountain pen. It, as with the pocket dosimeter, works on the principle similar to an ion chamber. The pocket chamber will indicate accumulated exposure to radiation at any time, but it requires a separate unit to read the device. As with the pocket dosimeter, it is calibrated in terms of milliroentgens and also requires recharging to reset the unit for repeated usages.

## Choice and Use of Instruments

The factors that affect the selection of a particular instrument for use in monitoring ionizing radiation include the directional dependence of the instrument, the response rate and range, the susceptibility of the instrument to the environmental interference, and the precision and accuracy of calibration as required by the user. Table 6.4.2 outlines generally the typical use of each of the devices discussed.

It will be necessary, however, to evaluate each individual setting to determine the specific instrument that would be appropriate for monitoring that environment.

Table 6.4.2

Types of detectors used
for various types of radiation.

| Type of Detector | Type of Radiation |
|---|---|
| Proportional or Scintillation Counter | Alpha |
| Geiger-Mueller Tube or Proportional Counter | Beta, Gamma |
| Ionization Chamber | X- and Gamma |
| Proportional Counter | $N_f$ |
| Proportional Counter | $N_t$ |